

SCRUMI: A Board Serious Virtual Game for Teaching the SCRUM Framework

Adler Diniz de Souza, Rodrigo Duarte Seabra

UNIFEI – Universidade Federal de Itajubá
Av. BPS, 1303, Pinheirinho, Instituto de Matemática e
Computação, CEP 37500-903 – Itajubá, MG
adlerunifei@gmail.com, rodrigo@unifei.edu.br

Juliano Marinho Ribeiro, Lucas E.da S. Rodrigues

UNIFEI – Universidade Federal de Itajubá
Av. BPS, 1303, Pinheirinho, Instituto de Matemática e
Computação, CEP 37500-903 – Itajubá, MG
juliano.ccounifei@gmail.com, lucasesr@gmail.com

Abstract—The use of serious games has emerged as a differentiated strategy to promote the teaching of essential concepts and techniques in several areas of knowledge. To contribute to the student's formation process in Software Project Management, this research presents the development and validation of an electronic board serious game, named SCRUMI, for teaching concepts inherent to the SCRUM framework. The evaluation of the proposed game was carried out according to some criteria such as usability, quality of questions and activities presentation, applicability and motivation. The main results showed that the game is presented as a good alternative to be explored in the classroom.

Keywords—serious game; software project management; SCRUMI; SCRUM framework; experimental research.

I. INTRODUCTION

Due to the accelerated expansion of the games industry, many researches have been developed aiming to generate knowledge about the engagement produced by this way of entertainment. In this context, the concept of gamification emerged, which concerns the use of games for various entertainment purposes [1]. In practice, gamification is based on the use of mechanics observed in games in contexts that are not games [2]. The use of gamification in the educational field represents a valuable tool to kindle the interest of students facing traditional methodologies, making learning more interesting [3, 4].

Thus, this research aims to introduce the development of an electronic serious game, using some mechanics observed in board games, which can possibly support the teaching of concepts inherent to the SCRUM framework. The educational game proposed is expected to help students understand related contents for learning the discipline of Software Project Management. Moreover, the study investigates, from the students' perspective, the potential benefits provided due to the experienced use of the proposed game in that discipline. The main contributions of the study are: (1) to help spread the idea of using educational serious games so as to increase their use in education; (2) to qualitatively evaluate the opinion of the students involved in the study, verifying if the game developed met the expectations, as well as its degree of acceptance and contribution to the teaching process.

According to Brewer [5], in corporations, a large number of projects failed due to reasons related to mismanagement, caused by the absence of trained project managers.

This reality makes it essential to improve the educational process. To make the traditional teaching more enjoyable and fill in the blanks as to the practical experience required, according to Von Wangenheim and Prikladnicki [6], there is an emerging demand for new ways of teaching to improve project management classes.

Thinking of how to solve this problem in the Project Management context, several studies were consistently reported in the literature feature games developed exclusively for the teaching practices of the SCRUM framework: PlayScrum [7], The Scrum Game [8], Scrum from Hell [9], Ballpoint game [10], Scrum Game [11], Scrum Simulation with LEGO Bricks [12] and SCRUMIA [6]. Another approach can be found in the recent research by [13].

II. THE SCRUMI GAME

SCRUMI is a board electronic game consisting of questions and answers, whose goal is to introduce key concepts of the SCRUM framework. In its development, we used the activities that are part of the Sprint life cycle in a software project. The platform used for the game design was Unity. The game consists of two kinds of questions: (1) multiple choice or (2) drag and drop. The questions were distributed into five phases, namely: **(i) Preparation**: this phase presents existing concepts in Sprint zero, ranging from the formulation of the Scrum team to the definition of the product initial backlog. The goal is to teach the player the actions carried out in the early stages of a project utilizing SCRUM before the Sprint planning meeting; **(ii) Analysis**: it covers activities related to the Sprint planning meeting. At this stage, among other activities, we address the procedures for the stories score present in the product backlog and the creation of the Sprint backlog, which is a decisive factor for starting Sprint; **(iii) Execution**: at this stage, the concepts related to Sprint are presented, focusing on product development. The activities of this game phase aim to teach the players what occurs during Sprint by discussing practical issues, namely: (1) the emergence of unforeseen events; (2) the division of tasks; (3) updating the Task Board; and, finally, (4) the burndown chart. **(iv) Monitoring and control**: it covers issues related to daily meetings and the Sprint review; **(v) Closing**: it contemplates the last two activities of SCRUM, before the Sprint closure, Review Meeting and Sprint Retrospective.

Figure 1 illustrates the activities of two phases of the life cycle and the game control panel. The players' progress

occurs when they hit the issue related to the activity being conducted in which he is on the board. The player who moves all the spaces within the stipulated time wins the game. Each issue presents an estimated response period of one minute, and the game total time consists of 44 minutes. The player who misses more than 10 questions or spends more than 30% of the estimated time accumulated during the match loses the game. The player who spends less time on issues will have less time spent accumulated and, consequently, will have more time available to answer the next questions.

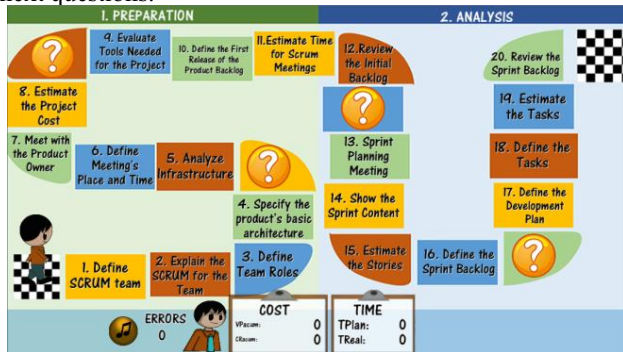


Figure 1. Activities of the preparation and analysis phases.

To facilitate monitoring the player status, there is a control panel (Figure 2) that allows the activation and/or monitoring of the following actions: (1) to enable/disable the sound by the button indicated in 1; (2) to show the number of player's errors indicated in 2; (3) to control the total time spent in the game indicated in 3; (4) time spent in current activity indicated in 4; (5) the clock (indicated in 5) display will flash in yellow when the player hits 50 seconds on the issue, and will flash in red when the time exceeds the estimated duration of 60 seconds for the issue.



Figure 2. SCRUMI game control panel.

After the players answer it, the game provides a feedback stating whether they hit or missed the issue, and if they spent more or less time than the estimated. Whenever a player misses a question, the player wastes the time spent on the issue because he/she does not advance any board space. If the player sets the question, he/she may: (1) save time for the next questions, if answered in less than 60 seconds; (2) waste time, otherwise. Since every project has a risk element involved, the spaces represented by an interrogation were added to the board. These spaces bring random events, positive or negative, to the time and cost of the project and are always related to phase activities of the life cycle in which the player is at the moment. The game can be accessed via the Internet at: <http://scrumigp.com>.

The game was evaluated for a Project Management class composed of 34 students who, after playing the SCRUMI, answered a questionnaire (developed based on the ISO/IEC 25010 standard) composed of 27 questions based on the

following attributes: (1) **usability**, (2) **quality of questions and activities presentation**, (3) **applicability**, (4) **motivation**. All the questions presented answers based on Likert scale, whose possibilities of response varied from 1 to 5, namely: 1 – strongly disagree, 2 – partially disagree, 3 – indifferent, 4 – partially agree and 5 – totally agree.

III. DATA ANALYSIS

For discussing the results, the following feedbacks will be considered: (1) positive – evaluated questions with answers 4 or 5; (2) indifferent – to the responses evaluated as 3; (3) negative – in the case of answers 1 or 2. The **usability** attribute had 77% of positively evaluation; 11% rated as indifferent and 12% disagreed with some question (Figure 3).

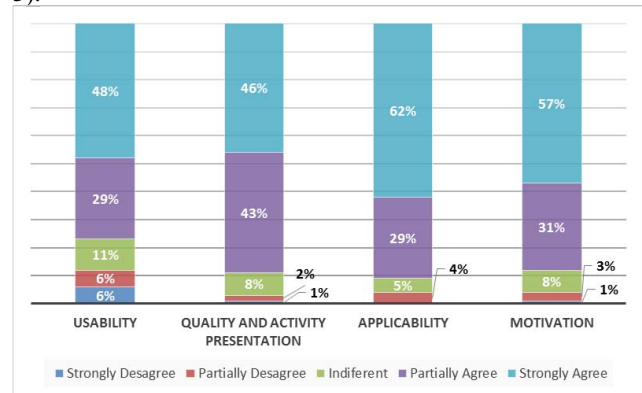


Figure 3. Activities of the preparation and analysis phases.

The **quality of questions and activities presentation** attribute had 89% of positive evaluation; 8% assessed some questions as indifferent and 3% of the students evaluated this attribute negatively (Figure 3). The **applicability** attribute showed an increase in positive feedbacks, adding up to a total of 91%. As for the negative results, only a few cases were observed in which some students partially disagreed with a question. Considering the **motivation** attribute, the percentage of positive ratings lay in the range of 88%. The percentage of indifferent opinions in this attribute was 8% and the negative answers amounted to 4% (Figure 3). In relation to the overall average, the students felt satisfied and had fun playing SCRUMI.

IV. FINAL CONSIDERATIONS

Based on the evaluation carried out from experienced usage of the SCRUMI game, we can notice that most of the students felt motivated to face the game issues, seeking to supplement the proposed challenges.

Considering these, among other results, the use of games and their mechanics could be more exploited in teaching Software Project Management and other disciplines. This statement can be proven by the positive results observed in the study, and it can be concluded that games add a differentiated and enjoyable experience, in addition to allowing better understanding of the content of the subjects worked on in the classroom. The authors would like to thank the financial support granted by FAPEMIG and CNPq.

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